WORKED SOLUTION

STEP I 2015, Question 10

Mechanics • Relative Velocity

STEP I 2015, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q10

The key idea

Rain hitting a surface is a flux: the amount per unit time is (number of drops per unit volume) $\times$ (area) $\times$ (component of the rain's velocity perpendicular to the surface). When the bus moves, work in the bus's frame, where only the horizontal component of the rain's velocity changes. Then, for the whole journey, multiply the rate by the journey time D/u and minimise over u. The modulus splits this into two cases, according to whether the bus is slower or faster than the horizontal motion of the rain.

Solution

The rate at which rain hits the bus

Model the rain as drops spread uniformly through the air, with n drops per unit volume, all moving with the same velocity. Let the bus have width b. Take north as the positive horizontal direction. Rain coming *from* the south moves northwards, so its velocity has

$$\text{horizontal component } v\cos\theta \text{ (north)}, \qquad \text{vertical component } v\sin\theta \text{ (down)}.$$

The rain has no east–west component, so the two long sides of the bus stay dry.

The case $u = 0$. In time δt, the drops that hit the roof are those in a slanting prism standing on the roof, of vertical height $v\sin\theta\,\delta t$. Its volume is $ab\,v\sin\theta\,\delta t$. The drops hitting the back (south face, area hb) fill a prism of horizontal depth $v\cos\theta\,\delta t$ behind it. The front, facing north, is sheltered. So the rate is

$$nb\,(h\,v\cos\theta + a\,v\sin\theta).$$

The case $u > 0$. In the frame of the bus, which is moving at a constant velocity, the rain has velocity

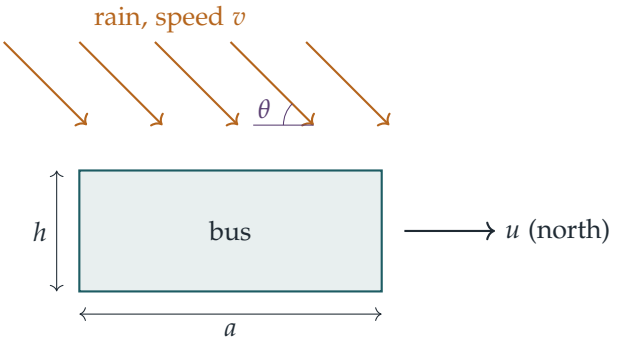
$$(v\cos\theta - u) \text{ north}, \qquad v\sin\theta \text{ down}.$$

The bus is at rest in this frame, so the argument above applies with $v\cos\theta$ replaced by $v\cos\theta - u$. The vertical component is unchanged, so the roof still collects at rate

$nab\,v\sin\theta$. If $v\cos\theta > u$, the rain still overtakes the bus and hits the back at rate $nhb\,(v\cos\theta - u)$. If $v\cos\theta < u$, the bus runs into the rain, which now hits the front at rate $nhb\,(u - v\cos\theta)$. In every case the total rate is

$$nb\Big(h\,|v\cos\theta - u| + a\,v\sin\theta\Big),$$

which is proportional to $h|v\cos\theta - u| + av\sin\theta$, as required.



Choosing the speed

Let the journey have length D. It takes time D/u, so the total rain collected is proportional to

$$R(u) = \frac{D}{u}\Big(h|v\cos\theta - u| + av\sin\theta\Big), \qquad 0 < u \le w.$$

Write $c = v\cos\theta$ and remove the modulus.

If $u \le c$:

$$R(u) = D\left(\frac{hv\cos\theta + av\sin\theta}{u} - h\right).$$

The numerator is positive, so R is strictly decreasing on $0 < u \le c$.

If $u \ge c$:

$$R(u) = D\left(h + \frac{v\,(a\sin\theta - h\cos\theta)}{u}\right).$$

Whether this increases or decreases depends only on the sign of $a\sin\theta - h\cos\theta$.

Ask yourself Why is going faster not always better?

Going faster shortens the journey, which always helps the roof: the roof's share is $Dav\sin\theta/u$. Once $u > c$, though, the bus runs into the rain, and the front's share $Dh(u-c)/u$ grows with u. The sign of $a\sin\theta - h\cos\theta$ tells us which effect wins.

The two formulae agree at $u = c$, so R is continuous. Now:

- If $w < v\cos\theta$, every allowed speed has $u \le c$, where R is decreasing. So the best choice is $u = w$.

- If $a\sin\theta > h\cos\theta$, then R is decreasing on $u \le c$ and also on $u \ge c$. So R is decreasing on the whole of $(0, w]$, and again the best choice is $u = w$.

This proves the first claim.

If $w > v\cos\theta$ and $a\sin\theta < h\cos\theta$: R decreases on $(0, c]$ and strictly increases on $[c, w]$. So the minimum is at

$$u = v\cos\theta,$$

which means driving at exactly the horizontal speed of the rain. The rain then falls vertically relative to the bus, so only the roof gets wet, and the total is $R(c) = Da\tan\theta$ (in our units).

If $a\sin\theta = h\cos\theta$: for $u \ge c$, $R(u) = Dh$, which does not depend on u. If $w \ge v\cos\theta$, then any speed between $v\cos\theta$ and w is equally good. Every speed below $v\cos\theta$ is worse. (If $w < v\cos\theta$, the first case applies and $u = w$.) Physically, beyond $u = c$, going faster saves exactly as much rain on the roof as it adds on the front. The condition says $\tan\theta = h/a$: seen from the road, the rain falls parallel to a diagonal of the bus's side.

The return journey

Now the bus travels south, into the rain. Its velocity is $-u$ in our northward convention, so the relative horizontal velocity of the rain is $v\cos\theta + u > 0$, and it always hits the front. The total is proportional to

$$R(u) = \frac{D}{u}\Big(h(v\cos\theta + u) + av\sin\theta\Big) = D\left(h + \frac{v\,(h\cos\theta + a\sin\theta)}{u}\right),$$

which is strictly decreasing in u. So on the way back the driver should go as fast as possible, $u = w$, whatever the values of a, h and θ.

Checks

- With $a = 1.3$, $h = 0.8$, $v = 2$, $\theta = 0.6$ and $u = 0.7$, the formula gives a rate of 2.229 (per unit drop density and width). A computer simulation of randomly placed drops crossing the bus's cross-section gave 2.232. The agreement was equally good for $u = 0$, for $u > v\cos\theta$ and for negative u (the return journey).
- Take $\theta \to \frac{1}{2}\pi$ (rain falling vertically). Then $a\sin\theta > h\cos\theta$, so we should drive at full speed. That makes sense: only the roof gets wet, and less time in the rain means less rain.
- In the equality case with $\tan\theta = h/a$, the total Dh is the same at $u = v\cos\theta$, $1.5v\cos\theta$ and $3v\cos\theta$, as predicted.

Common slips

- Minimising the *rate* instead of the *total*: the journey time D/u must multiply the rate.

- Forgetting that for $u > v\cos\theta$ the rain hits the front, not the back. This is the reason for the modulus.
- On the return journey, using $|v\cos\theta - u|$ again; the relative speed is now $v\cos\theta + u$.

Beyond STEP

The calculation "density $\times$ area $\times$ normal velocity" is the flux of a vector field through a surface. In vector calculus this becomes the surface integral $\iint \rho\,\mathbf{v}\cdot\mathbf{n}\,\mathrm{d}S$, which is the starting point for conservation laws in fluid dynamics and electromagnetism. Changing to the bus's frame is a Galilean transformation, the classical forerunner of relativity. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both relative motion and flux arguments of this kind.

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